Rakshith Suresh

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MS EE student who spent 10 months building a RISC-V matrix-multiply SoC from RTL to GDS on Sky130, ASAP7 and IHP, with gate-level simulation passing and an IHP SG13CMOS5L submission planned. Seeking Spring 2027 co-op and full-time hardware roles from May 2027.

Education

University of Southern California — Los Angeles, CA
Master of Science in Electrical Engineering, VLSI and Digital Design — *Aug 2025 – May 2027*
- Coursework: MOS VLSI Design (EE477), VLSI System Design (EE577a, EE577b), Computer Systems Organization (EE457)

Visvesvaraya Technological University — Bangalore, India
Bachelor of Engineering in Electronics and Communication Engineering — *Jan 2020 – May 2024*

Technical Skills

Verification: UVM, SVA, Formal Verification (SymbiYosys/Z3), Functional Coverage, VCS, Verilator, cocotb, uvm-python

HDL & Programming: SystemVerilog, Verilog, Python, C, C++, TCL, Perl

Physical Design: Floorplanning, Timing Closure, DRC/LVS/XOR, Cadence Innovus, Synopsys PrimeTime, OpenROAD, OpenLane 2, KLayout

Process Technologies & Tools: SkyWater 130nm, IHP SG13G2/SG13CMOS5L 130nm, GF 180nm, ASAP7 7nm, FPGA (Vivado, Artix-7)

Projects

Argo: RISC-V + Systolic Array SoC on IHP 130nm [GitHub] [GitLab] — Jan 2026 – Present
SystemVerilog, LibreLane, OpenROAD, Yosys, KLayout, Icarus – RTL to GDS on Sky130, ASAP7, IHP SG13G2/SG13CMOS5L
- **3 process nodes, RTL to GDS**: Submitted the 8x8 systolic array to the ChipFoundry CI2609 Sky130 shuttle, hardened the PicoRV32 SoC on ASAP7 7nm (500 MHz, 0 DRC), and reached clean DRC/LVS/antenna signoff on IHP SG13G2.
- **Argo SoC, 6.4 GMAC/s at 100 MHz typical**: Designed a PicoRV32 SoC with an 8x8 int8 systolic array, 2x4 KB IHP SRAM macros, SPI boot and UART on a 2x2 mm die with 68.7K cells, and met setup at the typical corner and hold at all PVT corners with a reset synchronizer and tuned post-route repair.
- **Signoff-clean GDS on IHP SG13CMOS5L**: Reached 0 DRC, density and antenna errors with an LVS match using only KLayout signoff, by building local LVS, fill and bond-pad fixes for a PDK with no Magic support, and contributed a golden design to Symbiotic EDA's CI flow.
- **Gate-level simulation, 2/2 programs and 40/40 checks**: Verified the routed netlist with typical-corner SDF simulation, booting each program over SPI (307-word image) and checking array outputs across 5 vector sets against a CPU reference model.

Custom ISA Hardware Accelerator [GitHub] — Jun 2026 – Present
SystemVerilog, Python, C, Verilator, Vivado – RISC-V processor with a custom hardware accelerator
- **27/27 tests passing**: Extended a RISC-V (RV32I) processor with 3 custom matrix-multiply instructions, validating every instruction against a Spike ISS reference model.
- **757 MHz, 0 DRC on ASAP7 7nm**: Closed setup and hold timing across 10,930 cells, then brought the design up on an Arty A7 FPGA, independently root-causing and resolving a board-level reset polarity issue during bring-up.

UVM Verification Environment for 8x8 Systolic Array [GitHub] — Apr 2026 – Present
uvm-python, cocotb, Python, Verilator, VCS – Self-checking UVM testbench for a matrix-multiply accelerator
- **1,337/1,337 checks passed, 0 mismatches**: Built a self-checking UVM environment (sequencer, driver, monitor, scoreboard) with a coverage plan closed across 5 regression seeds, using an independent reference model.
- Root-caused a VCS vs. Verilator discrepancy to a timescale LRM violation, restoring identical results across both tools.

Multi-Warp SIMT GPU Core in RV32I [GitHub] — Jun 2026 – Present
SystemVerilog, Yosys, OpenROAD, ASAP7 – 2-warp, 8-lane SIMT execution engine
- **34/34 tests passing, 500 MHz, 0 DRC**: Designed a round-robin-scheduled, 2-warp SIMT core (283 cells plus 8 macros, 2,863 μm², ASAP7 FF corner, +1,454 ps setup and +50 ps hold slack) and verified both threads against Spike ISS.

RV32I 5-Stage Pipeline CPU [GitHub] — May 2026
SystemVerilog, SymbiYosys, Z3 – Branch predictor, hazard unit, formal verification
- **27/27 tests, 5 formal properties**: Proved 5 SVA properties on the hazard unit with SymbiYosys and co-simulated against Spike ISS.

Custom Development Board — In Progress
PCB design – Learning board-level design
- Teaching myself to design my own development board for bringing up my chip.

Experience

Toyota Kirloskar Motor Private Limited — Bangalore, India
Electrical Engineering Intern — *Aug 2023 – Oct 2023*
- Diagnosed and resolved electrical faults across 12 production line systems by analyzing schematics and identifying root cause.

Leadership

- **Viterbi Graduate Mentor**, USC. **Member**, IEEE USC Student Branch. **Class Representative**, VTU (4 years). **Project Lead**, inter-state pitch competition, 3rd place finish.